

4.3–4.6 Study Guide

September 2026

Recommended Continued Practice (not for credit)

Find the local extrema of f and the intervals on which f is increasing or is decreasing, and sketch the graph of f .

1. $f(x) = 10x^3(x - 1)^2$

Find the local extrema of f on $[0, 2\pi]$ and the subintervals on which f is increasing or is decreasing. Sketch the graph of f .

2. $f(x) = \frac{1}{2}x - \sin x$

Find the local extrema of f , using the second derivative test whenever applicable. Find the intervals on which the graph of f is concave upward or is concave downward, and find the x -coordinates of the points of inflection. Sketch the graph of f .

3. $f(x) = \sqrt[5]{x} - 1$

4. $f(x) = \sqrt[3]{x^2} (3x + 10)$

Sketch the graph of a continuous function f that satisfies all of the stated conditions.

5. $f(0) = 3$; $f(-2) = f(2) = -4$; $f'(0)$ is undefined;
 $f'(-2) = f'(2) = 0$; $f'(x) > 0$ if $-2 < x < 0$ or $x > 2$;
 $f'(x) < 0$ if $x < -2$ or $0 < x < 2$

6. $f(0) = 1$; $f(2) = 3$; $f'(0) = f'(2) = 0$;
 $f'(x) < 0$ if $|x - 1| > 1$; $f'(x) > 0$ if $|x - 1| < 1$;
 $f''(x) > 0$ if $x < 1$; $f''(x) < 0$ if $x > 1$

7. $f(0) = 2$; $f(2) = f(-2) = 1$; $f'(0) = 0$;
 $f'(x) > 0$ if $x < 0$; $f'(x) < 0$ if $x > 0$;
 $f''(x) < 0$ if $|x| < 2$; $f''(x) > 0$ if $|x| > 2$

8. $f(1) = 4$; $f'(x) > 0$ if $x < 1$; $f'(x) < 0$ if $x > 1$;
 $f''(x) > 0$ for every $x \neq 1$

9. If n is an odd integer, then $f(n) = 1$ and $f'(n) = 0$; if n is an even integer, then $f(n) = 0$ and $f'(n)$ does not exist; if n is any integer, then

(a) $f'(x) > 0$ whenever $2n < x < 2n + 1$

(b) $f'(x) < 0$ whenever $2n - 1 < x < 2n$

(c) $f''(x) < 0$ whenever $2n < x < 2n + 2$

Find the extrema and points of inflection, and sketch the graph of f .

10. $f(x) = \frac{3x}{(x+8)^2}$

Simplify $f(x)$ and sketch the graph of f .

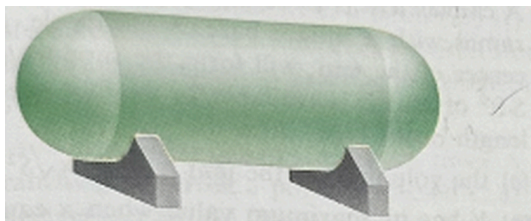
11. $f(x) = \frac{x-1}{1-x^2}$

Sketch the graph of f .

12. $f(x) = |x^3 + 1|$

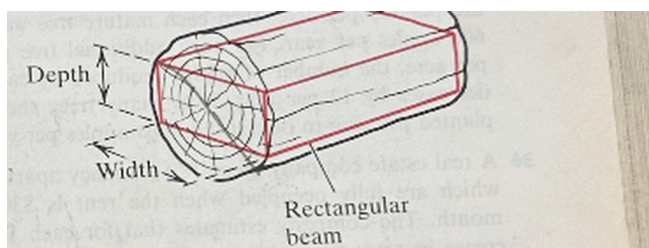
13. A builder intends to construct a storage shed having a volume of 900 ft^3 , a flat roof, and a rectangular base whose width is three-fourths the length. The cost per square foot of the materials is \$4 for the floor, \$6 for the sides, and \$3 for the roof. What dimensions will minimize the cost?

14. A steel storage tank for propane gas is to be constructed in the shape of a right circular cylinder with a hemisphere at each end (see figure). The construction cost per square foot for the end pieces is twice that for the cylindrical piece. If the desired capacity is $10\pi \text{ ft}^3$, what dimensions will minimize the cost of construction?

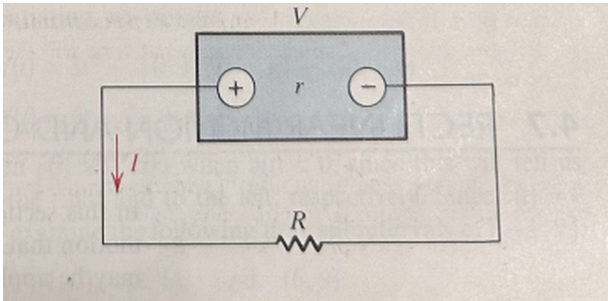


15. Find the point on the graph of $y = x^2 + 1$ that is closest to the point $(3, 1)$.

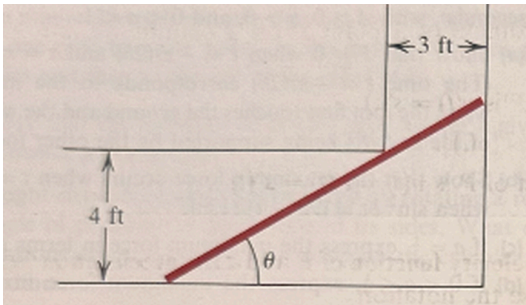
16. The strength of a rectangular beam is directly proportional to the product of its width and the square of the depth of a cross section. Find the dimensions of the strongest beam that can be cut from a cylindrical log of radius a (see figure).



17. A battery having fixed voltage V and fixed internal resistance r is connected to a circuit that has variable resistance R (see figure). By Ohm's law, the current I in the circuit is $I = V/(R + r)$. If the power output P is given by $P = I^2 R$, show that the maximum power occurs if $R = r$.



18. Two corridors 3 feet and 4 feet wide, respectively, meet at a right angle. Find the length of the longest nonbendable rod that can be carried horizontally around the corner, as shown in the figure. (Disregard the thickness of the rod.)



Tommy's Questions

These three are written by us. Each one pulls several ideas together at once, so show all of your work.

Tommy 1. Let $f(x) = 3x^5 - 20x^3 + 60x$.

- (a) Find $f'(x)$ and every critical number of f .
- (b) Find the intervals on which f is increasing and those on which f is decreasing, and find every local extremum of f .
- (c) Apply the second derivative test at each critical number found in part (a), and state what it tells you about that number.
- (d) Find the intervals on which the graph of f is concave upward and those on which it is concave

downward, find the x -coordinates of all points of inflection, and sketch the graph of f .

Tommy 2. Find the minimum value of

$$f(x) = (x^2 - 6x + 10)(x^2 - 6x + 13)(x^2 - 6x + 18).$$

Tommy 3. Find the minimum value of

$$f(x) = \frac{x^4 + 2x^3 + 5x^2 + 4x + 8}{x^2 + x + 2}.$$

Answer Key

Answers only for 1–18. Tommy's questions are worked in full.

1. Increasing on $(-\infty, \frac{3}{5}]$ and $[1, \infty)$; decreasing on $[\frac{3}{5}, 1]$. Local maximum $f(\frac{3}{5}) = \frac{216}{625} = 0.3456$; local minimum $f(1) = 0$. ($x = 0$ is a critical number but not an extremum: f' does not change sign there.)
2. Decreasing on $[0, \frac{\pi}{3}]$ and $[\frac{5\pi}{3}, 2\pi]$; increasing on $[\frac{\pi}{3}, \frac{5\pi}{3}]$. Local minimum $f(\frac{\pi}{3}) = \frac{\pi}{6} - \frac{\sqrt{3}}{2} \approx -0.342$; local maximum $f(\frac{5\pi}{3}) = \frac{5\pi}{6} + \frac{\sqrt{3}}{2} \approx 3.484$.
3. No local extrema; f is increasing on $(-\infty, \infty)$. Concave upward on $(-\infty, 0)$, concave downward on $(0, \infty)$. Point of inflection $(0, -1)$, where the graph has a vertical tangent line.
4. Local maximum $f(-\frac{4}{3}) = 4\sqrt[3]{6} \approx 7.27$; local minimum $f(0) = 0$ (the second derivative test does not apply at $x = 0$, since $f''(0)$ does not exist — use the first derivative test). Concave downward on $(-\infty, 0)$ and $(0, \frac{2}{3})$; concave upward on $(\frac{2}{3}, \infty)$. Point of inflection $(\frac{2}{3}, 4\sqrt[3]{12}) \approx (0.67, 9.16)$. The graph has a cusp at the origin.
5. Decreasing on $(-\infty, -2)$, increasing on $(-2, 0)$, decreasing on $(0, 2)$, increasing on $(2, \infty)$. Local minima $(-2, -4)$ and $(2, -4)$; local maximum $(0, 3)$. Because $f'(0)$ does not exist, the graph must come to a sharp point at $(0, 3)$ — a corner or a cusp, not a smooth peak.
6. Decreasing on $(-\infty, 0)$, increasing on $(0, 2)$, decreasing on $(2, \infty)$. Local minimum $(0, 1)$; local maximum $(2, 3)$. Concave upward on $(-\infty, 1)$, concave downward on $(1, \infty)$; point of inflection at $x = 1$ (its height is not determined by the conditions — any value between 1 and 3 works).
7. Increasing on $(-\infty, 0)$, decreasing on $(0, \infty)$; local maximum $(0, 2)$. Concave upward on $(-\infty, -2)$ and $(2, \infty)$, concave downward on $(-2, 2)$. Points of inflection $(-2, 1)$ and $(2, 1)$.
8. Increasing on $(-\infty, 1)$, decreasing on $(1, \infty)$; local maximum $(1, 4)$. The graph is concave upward on $(-\infty, 1)$ and on $(1, \infty)$. Because it is concave upward on *both* sides of the high point, f cannot be differentiable at $x = 1$: the graph must come to a corner (or cusp) at $(1, 4)$, not a smooth hump.
9. A repeating pattern of period 2. Each arch starts at $(2n, 0)$, rises to a smooth high point $(2n + 1, 1)$ with a horizontal tangent, and falls back to $(2n + 2, 0)$; it is concave downward the whole way. At every even integer two arches meet in a corner, so f' does not exist there. The graph looks like $y = |\sin \frac{\pi x}{2}|$.
10. Local maximum $(8, \frac{3}{32})$; no local minimum. Concave downward on $(-\infty, -8)$ and $(-8, 16)$; concave upward on $(16, \infty)$. Point of inflection $(16, \frac{1}{12})$. Vertical asymptote $x = -8$; horizontal asymptote $y = 0$.
11. $f(x) = \frac{-1}{x+1}$ for $x \neq 1$ and $x \neq -1$. So the graph is the graph of $y = -\frac{1}{x+1}$ with a *hole* at $(1, -\frac{1}{2})$. Vertical asymptote $x = -1$; horizontal asymptote $y = 0$.
12. The graph of $y = x^3 + 1$ with the piece below the x -axis (that is, $x < -1$) reflected up across the x -axis. Decreasing on $(-\infty, -1)$, increasing on $(-1, \infty)$; absolute minimum $(-1, 0)$, which is a corner. Concave upward on $(-\infty, -1)$, concave downward on $(-1, 0)$,

concave upward on $(0, \infty)$. Point of inflection $(0, 1)$. ($x = -1$ is *not* a point of inflection: the concavity changes there, but there is no tangent line.)

13. Length $= 2\sqrt[3]{300} \approx 13.39$ ft, width $= \frac{3}{2}\sqrt[3]{300} \approx 10.04$ ft, height $= \sqrt[3]{300} \approx 6.69$ ft. (Minimum cost $= 630\sqrt[3]{90} \approx \2823.29 .)
14. Radius $r = \frac{1}{2}\sqrt[3]{15} \approx 1.23$ ft; length of the cylindrical piece $L = 2\sqrt[3]{15} \approx 4.93$ ft. (Note $L = 4r$; the tank is $3\sqrt[3]{15} \approx 7.40$ ft long overall.)
15. The point $(1, 2)$; the distance is $\sqrt{5} \approx 2.236$.
16. Width $= \frac{2\sqrt{3}}{3}a \approx 1.155a$ and depth $= \frac{2\sqrt{6}}{3}a \approx 1.633a$; that is, depth $= \sqrt{2} \times$ width.
17. $P = \frac{V^2 R}{(R+r)^2}$, so $\frac{dP}{dR} = \frac{V^2(r-R)}{(R+r)^3}$, which is positive for $R < r$ and negative for $R > r$.
Hence P is largest at $R = r$, and the maximum power is $\frac{V^2}{4r}$.
18. $(3^{2/3} + 4^{2/3})^{3/2} \approx 9.87$ ft.

Tommy 1. $f(x) = 3x^5 - 20x^3 + 60x$.

(a)

$$f'(x) = 15x^4 - 60x^2 + 60 = 15(x^4 - 4x^2 + 4) = 15(x^2 - 2)^2.$$

f' is defined for every x , so the critical numbers are the zeros of f' : $x = \sqrt{2}$ and $x = -\sqrt{2}$.

- (b) $(x^2 - 2)^2$ is a square, so $f'(x) \geq 0$ for *every* x , with equality only at $x = \pm\sqrt{2}$. The derivative therefore never changes sign:

f is increasing on $(-\infty, \infty)$, and f has **no** local extrema.

Both critical numbers fail the first derivative test. A horizontal tangent line is not the same thing as a high or low point — at $(\pm\sqrt{2}, \pm 32\sqrt{2})$ the graph flattens out for an instant and then keeps climbing.

(c)

$$f''(x) = 60x^3 - 120x = 60x(x^2 - 2),$$

so $f''(\sqrt{2}) = 60\sqrt{2}(2 - 2) = 0$ and likewise $f''(-\sqrt{2}) = 0$. The second derivative test is **inconclusive at both critical numbers** — it gives no information at all. This is exactly the situation the test is not built for, and part (b) is what settles the question.

- (d) Factor: $f''(x) = 60x(x - \sqrt{2})(x + \sqrt{2})$. Sign chart:

interval	$(-\infty, -\sqrt{2})$	$(-\sqrt{2}, 0)$	$(0, \sqrt{2})$	$(\sqrt{2}, \infty)$
f''	—	+	—	+
graph	concave down	concave up	concave down	concave up

The concavity changes at all three zeros, so the points of inflection are

$$(-\sqrt{2}, -32\sqrt{2}), \quad (0, 0), \quad (\sqrt{2}, 32\sqrt{2}),$$

since $f(\sqrt{2}) = 12\sqrt{2} - 40\sqrt{2} + 60\sqrt{2} = 32\sqrt{2} \approx 45.25$ and f is an odd function.

Sketch. An odd, everywhere-increasing curve through the origin, running from $-\infty$ to $+\infty$, with horizontal tangents at $(-\sqrt{2}, -32\sqrt{2})$ and $(\sqrt{2}, 32\sqrt{2})$. Those two horizontal tangents are points of *inflection*, not extrema — which is the whole point of the problem.

Tommy 2. Answer: the minimum value is $\boxed{36}$, attained at $x = 3$.

The way in. Every one of the three factors starts with the same $x^2 - 6x$. Complete the square once and it appears in all three:

$$x^2 - 6x + 10 = (x - 3)^2 + 1, \quad x^2 - 6x + 13 = (x - 3)^2 + 4, \quad x^2 - 6x + 18 = (x - 3)^2 + 9.$$

So if we write $t = (x - 3)^2$, then

$$f(x) = (t + 1)(t + 4)(t + 9), \quad \text{where } t \geq 0.$$

Finish. Each of $t+1$, $t+4$, $t+9$ is positive and increasing for $t \geq 0$, so their product is increasing for $t \geq 0$. The product is therefore smallest at the smallest allowed t , namely $t = 0$, and $t = (x - 3)^2 = 0$ exactly when $x = 3$:

$$f(3) = (0 + 1)(0 + 4)(0 + 9) = 36.$$

No differentiation was needed at all.

What brute force costs. Expanding gives

$$f(x) = x^6 - 18x^5 + 149x^4 - 708x^3 + 2020x^2 - 3264x + 2340,$$

and then

$$f'(x) = 2(x - 3)(x^2 - 6x + 16)(3x^2 - 18x + 34).$$

The two quadratics have discriminants -28 and -84 , so neither has a real zero and $x = 3$ really is the only real critical number. Same answer — after a page of expansion and a quintic to factor.

The trap. Treating the three factors as if they compete with one another, and hunting for a compromise value of x . They do not compete: all three bottom out at the same place, $x = 3$, because all three are built on the same $(x - 3)^2$.

Tommy 3. Answer: the minimum value is $\boxed{4}$, attained at $x = 0$ and at $x = -1$.

The way in. Square the denominator and compare it with the numerator:

$$(x^2 + x + 2)^2 = x^4 + 2x^3 + 5x^2 + 4x + 4,$$

which is the numerator less 4. So writing $u = x^2 + x + 2$,

$$f(x) = \frac{u^2 + 4}{u} = u + \frac{4}{u}.$$

One substitution turns a quartic-over-quadratic into two terms.

What values can u take? Completing the square, $u = (x + \frac{1}{2})^2 + \frac{7}{4}$, so $u \geq \frac{7}{4}$, and every value $u \geq \frac{7}{4}$ is actually attained (as x runs over the reals, u runs continuously from $\frac{7}{4}$ up to ∞).

Minimize $g(u) = u + \frac{4}{u}$ **on** $[\frac{7}{4}, \infty)$.

$$g'(u) = 1 - \frac{4}{u^2} = \frac{u^2 - 4}{u^2},$$

which is negative for $\frac{7}{4} \leq u < 2$ and positive for $u > 2$. So g decreases and then increases, with its minimum at $u = 2$:

$$g(2) = 2 + \frac{4}{2} = 4.$$

Since $2 \geq \frac{7}{4}$, the value $u = 2$ is available. Solving $x^2 + x + 2 = 2$ gives $x(x + 1) = 0$, so $x = 0$ or $x = -1$, and indeed

$$f(0) = \frac{8}{2} = 4, \quad f(-1) = \frac{1 - 2 + 5 - 4 + 8}{1 - 1 + 2} = \frac{8}{2} = 4.$$

The two traps.

- (1) *Minimizing u instead of f .* The smallest u occurs at $x = -\frac{1}{2}$, and there $f(-\frac{1}{2}) = \frac{7}{4} + \frac{16}{7} = \frac{113}{28} \approx 4.036$, which is *larger* than 4. In fact $x = -\frac{1}{2}$ is a local *maximum* of f , sitting between the two minima at $x = -1$ and $x = 0$. Making the inside small does not make $u + \frac{4}{u}$ small.
- (2) *Forgetting to check that $u = 2$ is reachable.* The unconstrained minimum of $u + \frac{4}{u}$ is at $u = 2$, but that is only the answer because 2 lies in the range $[\frac{7}{4}, \infty)$. Had the range been $u \geq 3$, the minimum would have been at the endpoint instead.

What brute force costs. The quotient rule, after simplifying, gives

$$f'(x) = \frac{x(x+1)(2x+1)(x^2+x+4)}{(x^2+x+2)^2},$$

whose real zeros are $x = -1$, $x = -\frac{1}{2}$ and $x = 0$ — the same three points, after a great deal more algebra.

Where These Questions Came From

These questions come from Sections 4.3, 4.4, 4.5 and 4.6 of the textbook. If you got one wrong, look it up and try the problems around it — that is where the extra practice is.

Section 4.3

1 = 4.3 #7

2 = 4.3 #19

5 = 4.3 #35

Section 4.4

3 = 4.4 #9

4 = 4.4 #11

6 = 4.4 #31

7 = 4.4 #33

8 = 4.4 #34

9 = 4.4 #37

Section 4.5

10 = 4.5 #19

11 = 4.5 #27

12 = 4.5 #31

Section 4.6

13 = 4.6 #11

14 = 4.6 #17

15 = 4.6 #23

16 = 4.6 #25

17 = 4.6 #51

18 = 4.6 #53

Tommy's Questions

Tommy 1 — original, not in the textbook

Tommy 2 — original, not in the textbook

Tommy 3 — original, not in the textbook